SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Application-Based Questions (2 Questions)

Course Code:	ITA0609	**Course Title:**	Machine Learning
CO Assessed:	CO1 – Analyze (BL 3)	**Assessment:**	Assessment Tool 2 – Application Based Questions.
Weightage:	40% of CIA	**Total Marks:**	30 (3 Questions ×10 Mark)
CO1 Statement:	*To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)*		
Student Name:	B. Poojitha	**Reg. No.:**	192424374
Section / Batch:	SLOT A	**Date:**	13.7.26

Code of Conduct

*I **Navuluri Anjali / 192324137** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

Annexure A – Application-Based Questions

1. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors.
   Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

Design of a Machine Learning System for Health Analysis and Disease Prediction

Aim

To design a machine learning system that predicts diseases such as heart disease or diabetes using patient health data including age, blood pressure, glucose level, and lifestyle factors.

Problem Statement

Early prediction of diseases such as **heart disease** and **diabetes** helps doctors provide timely treatment and reduce complications. A machine learning system can analyze patient medical records and predict whether a patient is at risk based on various health parameters.

Objective

- Predict whether a patient has a disease.
- Analyze patient health records.
- Assist doctors in making early diagnoses.
- Improve healthcare decision-making.

Input Features

Feature	Description	Data Type
Age	Age of patient	Integer
Gender	Male/Female	Categorical
Blood Pressure	Blood pressure (mmHg)	Integer
Glucose Level	Blood sugar level (mg/dL)	Integer
BMI	Body Mass Index	Float
Cholesterol	Cholesterol level	Integer
Smoking	Yes/No	Categorical
Alcohol Consumption	Yes/No	Categorical
Physical Activity	Low/Medium/High	Categorical
Family History	Yes/No	Categorical
Disease	Heart Disease / Diabetes (Target Variable)	Binary

ID	Age	BP	Glucose	BMI	Smoking	Exercise	Family History	Disease
1	45	140	180	31	Yes	Low	Yes	Yes
2	30	120	90	24	No	High	No	No
3	55	150	200	34	Yes	Low	Yes	Yes
4	38	125	95	26	No	Medium	No	No
5	60	160	220	36	Yes	Low	Yes	Yes
6	41	135	160	29	No	Medium	Yes	Yes
7	29	118	88	23	No	High	No	No
8	50	145	190	32	Yes	Low	Yes	Yes
9	35	122	100	25	No	Medium	No	No
10	62	165	240	38	Yes	Low	Yes	Yes
11	44	138	170	30	Yes	Medium	Yes	Yes
12	31	119	92	22	No	High	No	No
13	58	155	210	35	Yes	Low	Yes	Yes
14	36	124	98	24	No	Medium	No	No
15	47	142	175	31	Yes	Low	Yes	Yes
16	28	116	85	22	No	High	No	No
17	52	148	195	33	Yes	Low	Yes	Yes
18	39	128	105	27	No	Medium	No	No
19	57	152	205	34	Yes	Low	Yes	Yes
20	33	121	94	23	No	High	No	No

Data Preprocessing

Data preprocessing improves data quality before training.

Step	**Purpose**
Remove Missing Values	Prevent incomplete records from affecting predictions
Remove Duplicate Records	Eliminate repeated patient entries
Encode Categorical Data	Convert Yes/No or Male/Female into numerical values
Feature Scaling	Normalize BP, Glucose, BMI, etc., to similar ranges
Handle Outliers	Reduce the influence of abnormal values

Feature Selection

Feature selection identifies the most important variables affecting disease prediction.

Feature	Importance
Age	High
Blood Pressure	High
Glucose Level	Very High
BMI	High
Smoking	Medium
Physical Activity	High
Family History	High
Cholesterol	High

Methods:

- Correlation Analysis
- Chi-Square Test
- Recursive Feature Elimination (RFE)
- Random Forest Feature Importance

Machine Learning Algorithms

Algorithm	Why It Is Suitable
Logistic Regression	Simple, interpretable, and effective for binary classification (Disease: Yes/No).
Decision Tree	Easy to visualize and understand clinical decision rules.
Random Forest	High accuracy, handles noisy data well, reduces overfitting, and provides feature importance.
Support Vector Machine (SVM)	Performs well on complex classification problems with clear decision boundaries.

Algorithm	Why It Is Suitable
XGBoost	High predictive performance, especially for structured healthcare datasets.

Random Forest Classifier

Justification:

- High accuracy.
- Handles both numerical and categorical features.
- Less prone to overfitting compared to a single decision tree.
- Provides feature importance, helping identify the most influential health factors.
- Works well even when some features are correlated.

Model Training Steps

1. Collect patient data.
2. Clean and preprocess the dataset.
3. Select relevant features.
4. Split data into training (80%) and testing (20%) sets.
5. Train the Random Forest model.
6. Validate the model using the test data.
7. Predict disease status for new patients.

Model Evaluation Metrics

Metric Purpose

Accuracy Percentage of correct predictions
Precision Measures how many predicted positive cases are actually positive

Recall Measures how many actual positive cases are correctly identified F1-Score

Balances precision and recall

Confusion Matrix Shows correct and incorrect classifications

Metric	Purpose
ROC-AUC	Evaluates the model's ability to distinguish between classes

Expected Output

Patient Prediction Patient 1

Disease Detected Patient 2

No Disease

Patient 3 Disease Detected

Patient 4 No Disease

Advantages

- Early disease detection.
- Faster clinical decision support.
- Reduces diagnostic errors.
- Helps monitor high-risk patients.
- Supports preventive healthcare strategies.

Limitations

- Requires high-quality and representative patient data.
- Predictions depend on the quality of training data.
- May not generalize well to populations very different from the training dataset.
- Should assist, not replace, medical professionals.

Conclusion

A machine learning system for health analysis can effectively predict diseases such as heart disease and diabetes by analyzing patient information including age, blood pressure, glucose level, BMI, and lifestyle factors. Using proper **data preprocessing**, **feature selection**, and a robust algorithm such as **Random Forest**, the system can achieve accurate and reliable predictions. Such systems support healthcare professionals in early diagnosis and informed clinical decision-making.

2. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Applying the Find-S Algorithm for Loan Approval Prediction

Aim

To apply the Find-S algorithm to derive the most specific hypothesis that is consistent with all positive loan approval examples.

Introduction

The **Find-S Algorithm** is one of the simplest supervised machine learning algorithms used in **concept learning**. It starts with the **most specific hypothesis** and gradually generalizes it only when positive training examples require it. Negative examples are ignored during hypothesis construction.

Given Dataset

Example	**Employment**	**Salary**	**Credit Score**	**Loan History**	**Loan Approved**
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Attribute Description

Attribute **Possible Values**
Employment Employed, Unemployed

Salary High, Medium, Low Credit

Score Poor, Good, Excellent

Attribute	**Possible Values**
Loan History	Clean, Bad
Target	Loan Approved (Yes/No)

Find-S Algorithm

Steps

1. Start with the most specific hypothesis:

$$h = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

2. Consider only **positive examples (Loan Approved = Yes).**
3. For each positive example:
   - If the hypothesis attribute is empty (∅), replace it with the example's value.
   - If the attribute value differs from the current hypothesis, replace it with **?** (meaning any value).
4. Ignore all negative examples.
5. The final hypothesis is the most specific hypothesis that covers all positive examples.

Initial Hypothesis

Step	**Hypothesis**
Initial	⟨∅, ∅, ∅, ∅⟩

Example 1 (Positive)

Example
(Employed, High, Good, Clean)

Update hypothesis.

Step	**Hypothesis**
H_1	⟨Employed, High, Good, Clean⟩

Example 2 (Positive)

Example
(Employed, Medium, Good, Clean)

Compare with current hypothesis.

Attribute	**H_1**	**Example**	**Result**
Employment	Employed	Employed	Same
Salary	High	Medium	?
Credit Score	Good	Good	Same
Loan History	Clean	Clean	Same

Updated hypothesis

Step	**Hypothesis**
H_2	⟨Employed, ?, Good, Clean⟩

Example 3 (Negative)

Example
(Unemployed, Low, Poor, Bad)

Loan Approved = **No** According

to Find-S,

Negative examples are ignored.

Hypothesis remains unchanged.

Step	**Hypothesis**
H_3	⟨Employed, ?, Good, Clean⟩

Example 4 (Positive)

Example
(Employed, Medium, Excellent, Clean)

Compare with current hypothesis.

Attribute	Current	Example	Result
Employment	Employed	Employed	Same
Salary ?		Medium	?
Credit Score	Good	Excellent	?
Loan History	Clean	Clean	Same

Updated hypothesis

Step	**Hypothesis**
H_4	⟨Employed, ?, ?, Clean⟩

Hypothesis Evolution Table

Step	Training Example	Hypothesis
Initial	—	⟨∅, ∅, ∅, ∅⟩
Example 1	(Employed, High, Good, Clean)	⟨Employed, High, Good, Clean⟩
Example 2	(Employed, Medium, Good, Clean)	⟨Employed, ?, Good, Clean⟩
Example 3	Negative Example	No Change
Example 4	(Employed, Medium, Excellent, Clean)	⟨Employed, ?, ?, Clean⟩

Final Hypothesis

The most specific hypothesis consistent with all positive examples is:

⟨Employed , ? , ? , Clean⟩

This means:

- Employment **must be Employed**
- Salary **can be any value**
- Credit Score **can be any value**
- Loan History **must be Clean**

Advantages

Advantage Description
Simple Easy to understand and implement Fast

Processes only positive examples

Efficient Produces the most specific consistent

hypothesis Suitable Works well for small concept-learning

problems

Limitations

Limitation Description
Ignores Negative Examples May miss important information from rejected cases

Assumes Noise-Free Data Sensitive to mislabeled positive examples

Limited Generalization Works only when positive examples adequately describe the concept Single

Hypothesis Does not maintain multiple candidate hypotheses

Conclusion

The Find-S algorithm starts with the most specific hypothesis and generalizes it using only the positive training examples. For the given loan approval dataset, the final hypothesis is ⟨**Employed, ?, ?, Clean**⟩, indicating that an applicant is likely to receive loan approval if they are **employed** and have a **clean loan history**, while salary level and credit score may vary among the positive examples.